Secret Manager

Secret Manager lets you store, manage, and access secrets as binary blobs or text strings. With the appropriate permissions, you can view the contents of the secret.

Secret Creation

To create new “Secret” in Secret Manager, use EDP Prime Request and provide these details:

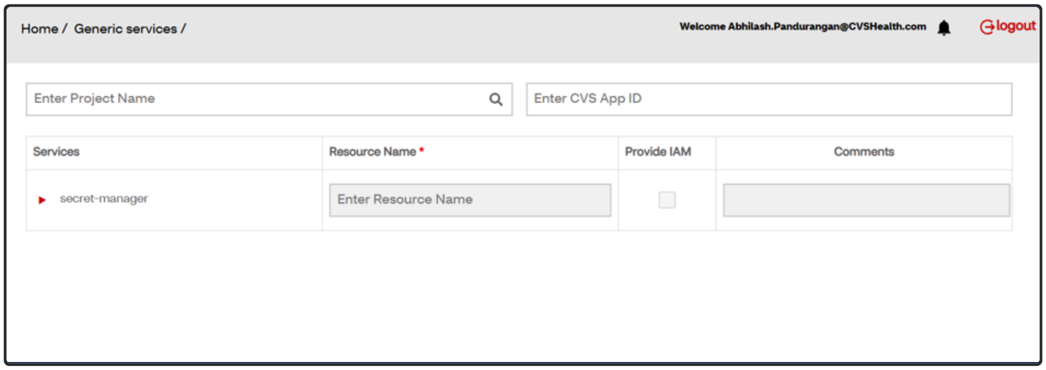
Using EDP Prime

> Secrets will be stored in GCP Compute Project only

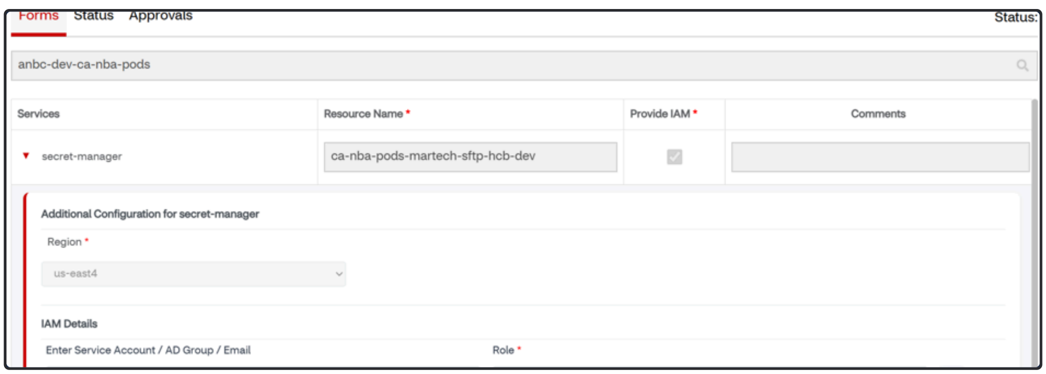
- Open Generic Services Request > Secret Manager
- Enter Project as your `compute project id` and your CVS App ID (`APM ID`) - We should create secrets in compute project only.
- Add Resource name should be in format - `<tenant>-<secret name you want>-<lob>-<env>` (Example: `ca-nba-pods-sftp-vendor-hcb-dev`).
- Click on `Provide IAM` :
  - Add roles with your email as `Secret Version Manager` , `Secret Accesor` (you need both to access and add using gcloud utility)
  - Add Compute SA as `Secret Accesor` (Ref: Secret Manager | Permissions on a secret)
- Once request is submitted, automated process will create within ~10mins.
- If access IAM is already provisioned, currently EDP prime ticket will be stuck in-progress. For any other issues related to EDP prime, reach out on Slack `#help-edp-prime-support`

> Service Account will not have access to Version manager role to update/add secret values. Only Secret accessor role will be provisioned for SA.

Request link: [CVS PRIME]



EDP Prime Self-Service process



Enter IAM permissions - Secret Accessor for Service account, Secret version manager for emails

> After creating new secret, update this list on confluence
>
> [Tenant-wise SID/ServiceAccount/Secret Manager/Azure SP Inventory]

Permissions on a secret

There are two types of roles for a specific secret.

- `Secret version manager` - Can add a new version (set value) of the secret. This is how the value of a secret can be set (version 1) or update (create a new version) - Access given to few people mentioned in ticket by IO team.

- `Secret accessor` - Can only read the secret but not set the value (SA and AD group by default)

Default IAM permissions for the secret varies by environments.

- Only few people specified in the justification box of intake form will have secret version manager role. The tenant AD group and resource SA will have secret accessor role.
- Only tenant resource service accounts have access to test and production secret manager and can set secret versions. If you want to add secret versions via the user interface, submit a GCP intake request to provide version manager access to existing secret manager for your newly created AD group or specific individuals. After a week, individual person access to test and production will be revoked. If you require access again, submit a fresh GCP intake request.

Best Practice

- NEVER print out the value of the secret in any case. Don't print out in composer dag or any code that logs are visible to everyone. By default everyone in a project can see all logs inside the project.
- Only use service account to access the value of the secret if the contents are not a JSON key for a service account
- If the secret is used for a Service Account Key, NO service account should be an accessor or version manager, only a select number of single users

Adding Secret

For those group of people in ticket with this role, you can add secret versions using `gcloud`

> Note: gcloud sdk needs to run in local and sso into your google account.
> Ref: [Install the gcloud CLI | Google Cloud SDK Documentation]

Before adding secret, make sure to set your project id in gcloud:

```
gcloud config set project PROJECT_ID
```

Make sure you have this role added to your email from IO: [Secret Manager Secret Version Manager] (`roles/secretmanager.secretVersionManager`)

To Add Secret, use this command:

```
gcloud secrets versions add <secret_id> --data-file="/path/to/file.txt"
```

Secret id would be full name of secret: `<tenant>-<secret name in ticket>-<env>`

Data file would be the file which contains the dictionary of secrets with “key” and “value” pairs in your local system.

Accessing Secret

All AD group and SA will have access to view secrets

```
gcloud secrets versions access version-id --secret=<mention secret name>
```

Python snippet to Access Secret

> Consider using Nexus package util - GCP CRUD helpers - Secret Manager Utilities

```python
import google.auth
from google.auth import impersonated_credentials
from google.cloud import secretmanager


def access_secret_version(
    resource_sa: str,
    project_id: str,
    secret_id: str,
    version_id: str
) -> secretmanager.AccessSecretVersionResponse:
    """
    Access the payload for the given secret version if one exists. The version
    can be a version number as a string (e.g. "5") or an alias (e.g. "latest").
    """

    credentials, project = google.auth.default()

    target_credentials = impersonated_credentials.Credentials(credentials,
                                                               target_principal=resource_sa,
                                                               target_scopes=[
                                                                   "https://www.googleapis.com/auth/cloud-platform"
                                                               ])

    # Create the Secret Manager client.
    client = secretmanager.SecretManagerServiceClient(credentials=target_credentials)

    # Build the resource name of the secret version.
    # project_id should be your compute project id on GCP
    name = f"projects/{project_id}/secrets/{secret_id}/versions/{version_id}"

    # Access the secret version.
    response = client.access_secret_version(request={"name": name})

```

```
    # WARNING: Do not print the secret in a production environment - this
    # snippet is showing how to access the secret material.
    payload = response.payload.data.decode("UTF-8")

    return payload
```

Using IO Intake

⚠ As of 08/28/2025, EDP Prime can be used to create new secrets which is self-service intake form. Following below is using IO intake process is not required.

IO Intake form Details

User Email	Provide your email
Aetna ID	Provide your A-ID
What do you need help with?	Request for gcp projects starting with anbc-* (existing tenants only)
Select a request type	Request new GCP resources or IAM for an existing tenant
Tenant	ca-nba-pods
Project	anbc-dev-ca-nba-pods (only one project per request) This should be compute project id (not shared project)
Type of resource	Secret manager secret (48 hour SLA)
Bucket location	
New instance or IAM only	IAM only
Name of instance	Note: The name will be automatically in format of `<tenant>-<secret name in ticket>-<env>` ℹ If you mention “sftp-vendor” in the ticket for dev environment, secret created will be `ca-nba-pods-sftp-vendor-hcb-dev`

	Note: “env” refers to environment without anbc so it will be hcb-dev for anbc-hcb-dev You don’t need to mention in this format.
Justification	Need secret to store SID/ password
Data Classification	confidential
(IAM only) Principals needed the IAM	If you mention project as dev, DEV SA will have access to it.
(IAM only) IAM Permission needed	• `Secret version manager` - Mention users who would need access to set the value of secret - This might be required for accessing/setting the value - You can request for 1-2 people who need this access. ***SA will not have access to this. Please request few members who need this not for everyone in AD group.*** • `Secret accessor` - Can only read the secret but not set the value - This by Default set to respective SA. *Service Account will not have access to Version manager role to update/add secret values. Only Secret accessor will be provisioned for SA.*

References

- Secret Manager Onboarding doc: https://github.aetna.com/abc-cloud/abc-gcp-onboarding/tree/master/secret-manager
- IO intake form: IO Intake
- Create and access a secret using Secret Manager | Secret Manager Documentation | Google Cloud